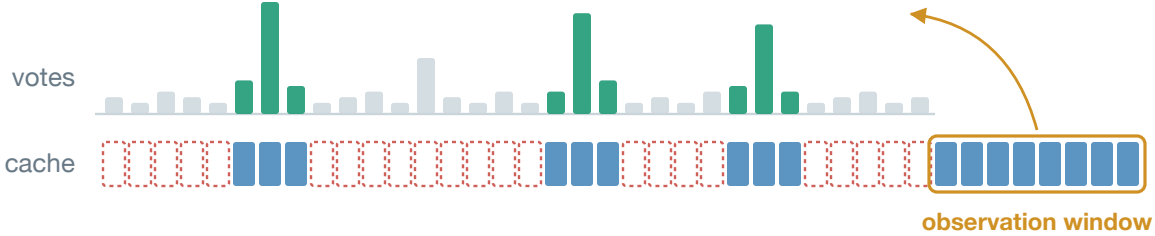
votes
cache
observation window